SKITII TRIALS PLATFORM

Product Story & Technical Specification

Architecture, session flows, weekly assessments, dashboard logic, and API contracts for the Skitii Clinical Trial app.

VERSION
1.0

DATE
July 2026

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SCOPE
React Native Tablet App + Trial Backend + Trial Admin Panel + Existing Auth Backend

1. Overview

Skitii Trials is a **clinical research version** of the Skitii product, built for running music-therapy trials at hospitals. The first customer is **Yenepoya Hospital**.

- The trial app is **almost identical to the Skitii product** in session flow (Pre Therapy → Baseline → Live Session → Post Therapy).
- The key difference is that the trial collects **extra patient clinical data** (cancer staging, ECOG, weekly psychological assessments) before and during the study.
- After the trial concludes, the data and learnings will feed back into the product.

Future: Other hospitals will run similar trials with **different patient intake forms**. The architecture must support per-hospital configurable patient schemas.

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2. Designed for Future Trials — Dynamic Architecture

The Skitii Trial platform is built from the ground up to support **multiple hospitals, multiple trials, and different clinical protocols** without requiring a new codebase for each.

2.1 What changes between trials

Element	Yenepoya v1 (current)	Future trials
Hospital / Institute	Yenepoya Hospital	Any hospital
Patient intake fields	Cancer staging, ECOG, diagnosis, chemo	Custom fields per trial
Trial duration	**4 weeks** (trial config; same for all patients)	Configurable per trial (`trialDurationWeeks`)
Weekly assessment forms	Dynamic per even week; steps differ per week	Configurable via `weeklyFormDefinitions` (**independent** of duration — any number of even-week entries)
Which even weeks exist	Seed example: **0, 2, 4** (may configure more, e.g. 7 form weeks)	Defined only in `weeklyFormDefinitions`
Form interval	**14 days** between form weeks	Configurable (`weeklyFormIntervalDays`)
Group types	Adaptive / Control	Configurable per trial
Number of shifts per day	3	Configurable
Shift time windows	Shift 1: 4AM–12PM, Shift 2: 12PM–8PM, Shift 3: 8PM–3:59AM	Configurable per trial

2.2 How we make it dynamic

1. `institutionId` + `trialId` scoping

Every record in the Trial DB is scoped to both `institutionId` and `trialId`. This means multiple hospitals can run different trials simultaneously on the same backend, with data fully isolated.

```
{
  "institutionId":   "ObjectId(...)",
  "trialId":         "skitii-trial-v1",
  "registrationNo":  "SKITII-2026-001"
}
```

2. Versioned patient intake schema

The patient intake form fields are not hardcoded. Each trial has a `patientSchema` document in the DB that defines which fields are collected, their types, labels, and validation rules. The Trial Admin Panel allows configuring this schema per trial without code changes.

```
{
  "trialId": "skitii-trial-v1",
  "patientSchema": [
    { "key": "diagnosisStage", "label": "Diagnosis Stage",         "type": "dropdown", "required":
true,  "options": ["Stage 1", "Stage 2", "Stage 3", "Stage 4"] },
    { "key": "treatmentType",  "label": "Treatment Type",          "type": "dropdown", "required":
true,  "options": ["Radiotherapy", "Chemotherapy", "Both"] },
    { "key": "cancerSite",     "label": "Cancer Site",             "type": "text",     "required":
true },
    { "key": "ecogScore",      "label": "ECOG Performance Status", "type": "dropdown", "required":
true,  "options": ["0", "1", "2", "3", "4"] },
    { "key": "surgicalType",   "label": "Surgical Type",           "type": "dropdown", "required":
true,  "options": ["Pre-Operative", "Post-Operative"] },
    { "key": "languagePref",   "label": "Language Preference",     "type": "dropdown", "required":
true,  "options": ["Kannada", "English", "Hindi", "Malayalam"] },
    { "key": "emergencyNumber","label": "Emergency Number",        "type": "phone",    "required":
true }
  ]
}
```

The React Native app and Admin Panel dynamically render the patient enrollment form from this schema — no app update needed when a new hospital has different fields.

3. Dynamic weekly form definitions (same pattern as patient schema)

Weekly assessment steps are **not hardcoded**. Each trial defines which **even weeks** exist and which steps each week includes.

> **`trialDurationWeeks` and `weeklyFormDefinitions` are independent.**
>
> - Duration (Yenepoya = **4**) → `completionDate` = enrollment + **(4 × 7 − 1) = 27** days; sessions allowed **through that day inclusive**, blocked from the next day.
> - Form list = whatever is in `weeklyFormDefinitions` (e.g. **3** or **7** even-week entries). Forms may still be submitted **after** `completionDate` if that week's window is open.

Config	Controls
`trialDurationWeeks`	When sessions stop (`completionDate`)
`weeklyFormDefinitions`	Which even-week forms exist and their steps (any length)

Yenepoya **seed example** (default forms — not a hard limit):

```
{
  "trialId": "skitii-trial-v1",
  "trialDurationWeeks": 4,
  "weeklyFormIntervalDays": 14,
  "weeklyFormDefinitions": [
    { "week": 0, "label": "Week 0", "steps": ["vitals", "gad7_phq9", "hrv"] },
    { "week": 2, "label": "Week 2", "steps": ["vitals", "gad7_phq9", "nccn_distress", "hads",
"qlq_c30", "qlq_hn35", "hrv"] },
    { "week": 4, "label": "Week 4", "steps": ["vitals", "gad7_phq9", "nccn_distress", "hads",
"qlq_c30", "qlq_hn35", "treatment_adherence", "satisfaction_survey", "hrv"] }
  ]
}
```

> Step keys and field-level detail: see **Weekly Clinical Review Forms** doc. `satisfaction_survey` is **intervention-group only**. `treatment_adherence` on Week **4** in the seed example. Progress = **Step X of N** for that week's configured steps.

4. Completion date vs sessions vs weekly forms (important)

After `completionDate` (next calendar day onward)	Allowed?
Start / restart a therapy **session**	**No** — `TRIAL_COMPLETED`
Submit a **weekly form** in `weeklyFormDefinitions` whose form window is still open (including weeks past duration, e.g. Week 6 / 8 if configured)	**Yes**

Example: duration **3** weeks → completion = enrollment + **20** days; duration **4** (Yenepoya) → + **27** days. If `weeklyFormDefinitions` has **7** form weeks, the patient can still fill those after completion while each window is open — sessions stay blocked.

5. Configurable trial parameters

Trial-level configuration (stored per `trialId`) controls all protocol parameters:

Parameter	Description
`trialDurationWeeks`	Session-period length for **all patients** (Yenepoya = **4**). Does **not** limit how many form weeks may be configured
`shiftsPerDay`	Number of session shifts per day (currently 3 for Yenepoya)
`shiftWindows`	Start/end times for each shift — **differs per trial**
`weeklyFormDefinitions`	Even-week form list + steps per week — **any number** of entries (independent of `trialDurationWeeks`)
`weeklyFormIntervalDays`	Gap between form week numbers (Yenepoya = **14** — Week 0, 2, 4…)
`groupTypes`	List of patient groups (e.g. Adaptive, Control)
`formReminderDaysBeforeExpiry`	Days before `windowEnd` to send form reminder (Yenepoya = **1**). Skip if form already Done. No Week 0 reminder

6. Trial Admin Panel as the control plane

The Trial Admin Panel is the central place where Skitii team configures each new trial. To onboard a new hospital:

1. Create an `Institution` record in Auth Backend.
2. Create a `Trial` record with all parameters, `patientSchema`, and `weeklyFormDefinitions`.
3. Assign the institution to the trial.
4. The mobile app and admin panel automatically work for that institution — no code deployment needed.

2.3 What stays the same across all trials

- Session flow: Pre Therapy → Baseline → Live Therapy → Post Therapy
- BLE integration with any compatible heart rate device
- HRV computation (RMSSD, SDNN, pNN50) on-device (weekly HRV = **5 minutes** when configured)
- Clinical day boundary concept (configurable start time, e.g. 4 AM – 3:59 AM)
- Weekly form model: Week 0 mandatory (submit before or on enrollment date; only Week 0 + enrollment date gate sessions); even weeks; **7-day** form windows on a 14-day week-number cadence; forms may continue after completion while sessions do not
- Auth mechanism (existing Auth Backend + JWT)
- Device PIN mechanism

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3. Architecture

2.1 Systems involved

System	Role
Existing Skitii Auth Backend	Email OTP login, JWT tokens, Institution identity
New Trial Backend	All trial domain data (patients, sessions, weekly reports, compliance)
New Trial DB (MongoDB)	Separate from product DB
New Trial Admin Panel	Admin interface for managing trial data, patients, and reports
Existing Skitii Admin Frontend	Admin adds `controlPatientMusic` flag to music assets
Trial React Native App	Tablet-only (landscape), Yenepoya institute
BLE Heart Rate Device	Any compatible BLE HR monitor used during sessions and weekly HRV

2.2 Auth flow

1. Trial app calls **Existing Auth Backend** → email OTP → JWT.
2. JWT is passed to **Trial Backend** for all domain API calls.
3. Trial Backend validates JWT and extracts `institutionId` to scope all data.
4. Yenepoya is registered as an Institution with `product: "skitii-trial"` in the Auth DB.

2.3 Multi-device

- One institute (Yenepoya) logs in on **multiple tablets**.
- PIN is **device-local** (stored in device Keychain/Keystore only).
- Each tablet creates its own PIN after OTP login.
- Trial Backend must support concurrent writes from multiple tablets under the same `institutionId`.
- Device restrictions follow the **same mechanism as the Skitii product** — no separate implementation for the Trial app.

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4. Trial Structure

3.1 Trial duration

- Duration is set once on the **trial configuration** (`trialDurationWeeks`) — **same for every patient**. Not chosen at enrollment.
- Formula: `completionDate` = `enrollmentDate + (trialDurationWeeks × 7 − 1)` days.
  - Yenepoya = **4** → **+ 27 days** (e.g. 17 Jul → **13 Aug**).
  - Example: **3** weeks → **+ 20 days**.
- Sessions are allowed **through** `completionDate` **inclusive**; blocked from the **next calendar day** (`TRIAL_COMPLETED`).
- Enrollment spanning month boundaries uses normal date arithmetic — no special handling.
- After the session period ends:
  - **Sessions:** cannot start. Backend returns `TRIAL_COMPLETED`; Start button disabled.

- **Weekly forms:** still allowed for **any** week in `weeklyFormDefinitions` whose form window is open (may be more form weeks than duration weeks).

- There is **no restriction on the number of sessions** within the trial period (through `completionDate`).

3.2 Weekly form calendar

- Forms exist only for **even study weeks** listed in `weeklyFormDefinitions` (any count — e.g. 3 or 7 entries).
- Week numbers advance by **14** (`weeklyFormIntervalDays: 14`).
- Each even-week form (Week 2+) has a fixed **7-day** window:
  - `windowStart` = `enrollmentDate + (studyWeek × 7)` days
  - `windowEnd` = `windowStart + 6` days (inclusive)
- Steps per week are **dynamic**.
- **Independent of duration:** configuring Week 6 / 8 / … does not require raising `trialDurationWeeks`; those forms open later (often with **no sessions**) and remain submittable while their windows are open.

Week 0 (special):

- May be submitted **before or on** the enrollment date. On enrollment day, must be done **before Shift 1 is within the 30-minute expiry rule** (if not already submitted earlier).
- Once Week 0 is submitted → patient is **view-only** (including enrollment date and group type). Set the correct enrollment date **before** an early Week 0 submit.
- **Sessions** require: Week 0 submitted **and** clinical date ≥ `enrollmentDate` (no sessions before enrollment day) **and** through `completionDate` inclusive.
- If Week 0 is still missing when Shift 1 hits the 30-min rule on enrollment day → institute **manually edits** enrollment date to the next day. Backend does **not** auto-change it. Sessions stay blocked until Week 0 is submitted.
- After Week 0: sessions continue through `completionDate` even if later weekly forms are pending or missed.
- **No form-reminder email** for Week 0.

Week 2 / 4 / 6…:

- Open on `enrollmentDate + (week × 7)` ; close 7 days later (inclusive).
- Form windows often begin **after** `completionDate` (e.g. Week 4+) — forms can be submitted; sessions cannot.
- Missed = window expired without `fully_completed` ; next configured even week still opens on its schedule.

Example — Yenepoya 4-week trial

Patient enrolled **Friday 17 Jul 2026**, `completionDate` = **13 Aug 2026** (enrollment + 27 days).
Illustrative form list Week **0, 2, 4, 6, 8** (seed may be 0/2/4 only — more weeks optional).

Study week	Sessions	Weekly form window
Week 0	From create through enrollment day	**Before or on 17 Jul (Fri) 2026**; on that day before Shift 1 30-min rule if not done earlier
Week 2	24 Jul (Fri) – 30 Jul (Thu) 2026	**31 Jul (Fri) – 6 Aug (Thu) 2026**
Week 4	7 Aug (Fri) – 13 Aug (Thu) 2026	**14 Aug (Fri) – 20 Aug (Thu) 2026**
Week 6	No Sessions	**28 Aug (Fri) – 3 Sep (Thu) 2026**
Week 8	No Sessions	**11 Sep (Fri) – 17 Sep (Thu) 2026**

> Sessions allowed through **13 Aug** inclusive; blocked from **14 Aug** (`TRIAL_COMPLETED`). Week 4 / 6 / 8 forms open after that — submit OK while each window is open. Reminder for Week 2 = **5 Aug** only (day before expiry), and only if not already Done.

If enrolled Tuesday 21 Jul 2026 (4-week, `completionDate` = 17 Aug 2026):

- Week 0: may submit **before or on 21 Jul (Tue)**; sessions only from **21 Jul** (Shift 1 30-min rule if submitting that day).
- Week 2 form: **4 Aug (Tue) – 10 Aug (Mon)** (reminder **9 Aug** if still pending).
- Week 4 form: **18 Aug (Tue) – 24 Aug (Mon)** (after sessions end).

Backend → frontend (per patient):

1. **Dashboard button label** — e.g. `Week 0` , `Week 2` , `Week 4`
2. **Steps to render** — from `weeklyFormDefinitions` for that week
3. **Form list** — `availableWeeks` = weeks from `weeklyFormDefinitions` (seed example `[0, 2, 4]` ; not derived from `trialDurationWeeks`)

Example labels: Patient A → Week 0 · Patient B → Week 2 · Patient C → Week 4.

3.3 Cron jobs & Email notifications

Three independent cron jobs run on the Trial Backend:

Cron 1 — Daily missed session report (runs at 4:00 AM every day)

Runs immediately after Shift 3 window closes (3:59 AM). If any sessions were missed for that clinical day, an email is sent to **both Skitii admin and the institute's login email**.

Email content:

- Total missed session count for the day

- Detailed list per patient: patient name + which shift(s) were missed

> No email is sent if there are zero missed sessions for that day.

Cron 2 — Weekly form reminder (day before window expiry)

For each open even-week form window (Week 2+ only — **not Week 0**):

- Send reminder on `windowEnd − formReminderDaysBeforeExpiry` (Yenepoya = **1** → the calendar day before expiry).
- Example — Week 2 window **31 Jul – 6 Aug**: reminder on **5 Aug only**.
- **Skip** patients whose form for that week is already **Done** (`fully_completed`).
- Do **not** send on other days of the window (not daily; not Thu/Fri/Sat sets).

Email goes to the **institute's login email** and lists patients with a pending form for that week whose reminder day is today.

Cron 3 — Weekly Form PIN rotation (every 14 days)

Generates a new 6-digit PIN every **14 days** (same cadence as `weeklyFormIntervalDays`). Old PIN is deactivated; admin informs institute by SMS/call.

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5. Session Shifts

4.1 Three daily shifts

Every patient must complete **3 sessions per day**, one per shift.

> **Shift time windows are configurable per trial** and set in the Trial Admin Panel. The windows below apply to **Yenepoya v1** only.

Shift	Window (Yenepoya v1)	Day attribution
Shift 1	4:00 AM → 12:00 PM	Same calendar day
Shift 2	12:00 PM → 8:00 PM	Same calendar day
Shift 3	8:00 PM → 3:59 AM (next day)	The day Shift 3 **started** (e.g. session at 2AM on July 18 = Shift 3 of July 17)

> A clinical "day" runs from the **shift 1 start time to shift 1 start time** of the next calendar day (for Yenepoya: 4:00 AM to 3:59 AM).

4.2 Shift ordering rules

1. Shifts must be completed in **strict order**: Shift 1 → Shift 2 → Shift 3.
2. **Exception:** If Shift 1 is missed (window closed), Shift 2 is still **allowed**.
3. A session that was **missed** in its shift **cannot be started in a later shift**.
4. Backend enforces all shift ordering and time-window validation.

4.3 Missed session

- A session is **missed** if:
  - Its shift window has closed, AND
  - No completed Post Therapy form was submitted for that shift.
- Missed sessions count toward **compliance statistics**.

4.4 Session status & termination (same as Skitii product)

Trial sessions use the **same status / terminationReason model** as the Skitii product:

Action	`status`	`terminationReason`
Session in progress	`ACTIVE`	—
Stop button	`INCOMPLETE`	`USER_ENDED`
Auto-end (~20 min timer)	`COMPLETED`	`SESSION_MAX_REACHED`
Backend timeout (30-min safety job)	`AUTO_TERMINATED`	`TIMEOUT`
New session started on same device	`AUTO_TERMINATED`	`AUTO_TERMINATED_ON_NEW_SESSION`

- Sessions **auto-end after ~20 minutes** → `COMPLETED` / `SESSION_MAX_REACHED`.
- A background job runs every **30 minutes** as a safety net — if the session is still `ACTIVE` past expiry, it is set to `AUTO_TERMINATED` / `TIMEOUT`.

> **Why shift window expiry during a live session is not possible:** The 30-minute minimum check (at Start + safety-net after Pre Therapy form) ensures ≥ 30 min remain before Baseline begins. Since a session auto-ends in ≤ 20 min, the shift window mathematically cannot close during an active session.

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6. Session Flow

Each session follows a linear 3-step flow with a stepper (Pre Therapy → Therapy Session → Post Therapy):

```
Dashboard (tap Action chip)
    ↓
POST /sessions/start — validates: duplicate session + 30-min minimum check [Point 1]
    ↓ (on success)
Pre Therapy Form
    ↓
POST /sessions/validate — re-validates: 30-min minimum check [Point 2 — safety net]
    ↓ (on success)
Baseline Measurement (live HR from BLE heart rate device)
    ↓
Live Therapy Session (BLE + music)
    ↓
Post Therapy Form
```

```
    ↓
Dashboard (Action chip → Done / Restart / Resume)
```

5.1 Pre Therapy Form

Collected before every session:

Vital Signs

- Blood Pressure Systolic* (mmHg)
- Blood Pressure Diastolic* (mmHg)
- Baseline Heart Rate (auto-filled from live Polar reading, read-only)

Patient Assessment

- Current Pain Level* (slider 0–10)
- Energy Level* (Good / Fair / Poor)
- Comorbidities Alert (optional text)
- Medication Changes since last session (optional text)

Session Setup

- Headphone Type* (Over-Ear / In-Ear)
- Posture* (Seated / Supine)
- Quiet Room Verified* (Yes / No)
- Session Environment Observations (optional text)

CTA: **Start Session** (disabled until required fields valid + Polar HR available)

5.2 Baseline Measurement

- Countdown timer — default **30 seconds**; **frontend-configurable** (same as Skitii product). Backend stores the recorded `durationSec`.
- Live HR + SD of HR from BLE heart rate device displayed.
- Heart rate waveform shown.
- "Please remain still and relaxed" instruction.
- Auto-advances to Live Session when timer completes.
- Stores: `startedAt`, `endedAt`, `durationSec`, `readings[{hr, timestamp}]`, `meanHr`, `sdHr`.

5.3 Live Therapy Session

For Adaptive patients — same as Skitii product:

- Music selected by HRV/HR-tier algorithm.
- Music ranking updated per shift based on RMSSD change.
- Shifts cycle with HRV rolling window.

For Control patients — different:

- Music pool = tracks flagged as `controlPatientMusic: true` in Skitii Admin.
- On session start: one random track from the pool plays continuously until session end.

- If patient changes music: another random track from the pool plays continuously until session end.
- Patient can change music anytime (unlimited changes).
- **Ranking is NOT updated** for control patients.
- Session readings store **multiple tracks per shift** (one entry per music change): `[{ trackId, startedAt, endedAt }, ...]` per shift.

Live metrics displayed:

- Device HR (bpm), HRV RMSSD, SD HR, AVG RR (150s), PNN50, SDNN, Artifact %
- Shift number, HRV tier, time remaining, session timer, REC indicator
- Patient name, LIVE badge, Polar battery

STOP button: Marks session as `INCOMPLETE` with `terminationReason: USER_ENDED`. Backend and frontend handle resume/restart logic (see Section 8).

5.4 Post Therapy Form

Collected after every session (or after Stop):

Vital Signs

- Blood Pressure Systolic* / Diastolic* (mmHg)
- Baseline Heart Rate (from pre-session/baseline, read-only display)

Patient Assessment

- Current Pain Level* (slider 0–10)
- Energy Level* (Good / Fair / Poor)

Observations

- Adverse Events* (toggle No/Yes + notes text if Yes)
- Technical Issues* (toggle No/Yes + notes text if Yes)

Data Summary

- RMSSD Change* (auto-filled from session, not manually editable)
- Average RMSSD* (auto-filled from session, not manually editable)
- Session Notes (optional text)

CTA: **Complete Session** (disabled until required fields valid)

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7. Weekly Assessment Flow (Dynamic Steps)

Triggered from Dashboard **Weekly → [Week N]** (button label comes from backend).

Steps are **not fixed**. The backend returns which steps to show for the current even week (from trial `weeklyFormDefinitions`). Yenepoya catalogue (illustrative — field detail in Weekly Forms doc):

Study week	Example steps
Week 0	Vitals → GAD-7 & PHQ-9 → 5-min HRV
Week 2	Vitals → GAD-7 & PHQ-9 → NCCN → HADS → QLQ-C30 → QLQ-H&N35 → 5-min HRV
Week 4	… + Treatment Adherence + Satisfaction Survey (intervention only) → HRV

```
Dashboard (tap Weekly button — label from backend, e.g. "Week 2")
    ↓
PIN entry → verify-pin → short-lived token
    ↓
Render only the steps configured for that week (Step X of N)
    ↓
Non-HRV steps submitted together (when present)
    ↓
HRV step (if configured) — 5-Minute Resting HRV, submitted separately
    ↓
Dashboard Weekly chip → Done
```

6.1 Available step types (configured per week)

When a step is included in that week's definition, fields apply as documented in **Weekly Clinical Review Forms**. Current Yenepoya step keys:

Step key	Display name	Notes
`vitals`	Vital Signs	Weekly Clinical Review: BP, SpO_2, ECOG + past history + meds + investigations — **no** SpO_2/BP min–max ranges
`gad7_phq9`	GAD-7 & PHQ-9	Combined step; totals on device + backend
`nccn_distress`	NCCN Distress Thermometer	Thermometer + problem checklist
`hads`	HADS-A & HADS-D	Validated option order
`qlq_c30`	QLQ-C30	EORTC v3 items 1–30
`qlq_hn35`	QLQ-H&N35	Module items 1–35
`treatment_adherence`	Treatment Adherence	Yenepoya: Week **4**; face-to-face
`satisfaction_survey`	Satisfaction Survey	Intervention group only (Week 4)
`hrv`	5-Minute Resting HRV	Separate API submit when configured

Non-HRV steps that are active for the week are submitted together in a single API call.

6.2 5-Minute Resting HRV (when configured for that week)

- Triggered after non-HRV steps are submitted (if any), or as a standalone step if it is the only remaining step.
- **5-minute** recording; patient must remain still with BLE heart rate device connected.
- Live UI same pattern as session **baseline** (HR + SD of HR + waveform).
- Payload shape = **baseline fields** + HRV metrics (no separate `heartRate` — use `readings` + `meanHr`):
  - Like baseline: `startedAt`, `endedAt`, `durationSec` (**300**), `meanHr`, `sdHr`, `readings[{ hr, timestamp }]`
  - Plus: `rmssd` (ms), `sdnn` (ms), `pnn50` (%)
- When the timer hits **5:00**, the frontend **automatically** posts to `POST /weekly-assessments/:id/hrv` (no manual Submit).
- On success → **Weekly Form Submitted Successfully!** screen → **Return to Dashboard**.
- Weekly assessment is `fully_completed` only after that HRV auto-post succeeds.
- Dashboard Weekly chip turns **Done** only when the week is fully completed.

6.3 Week 0 rules

- Week 0 may be submitted **before or on** the enrollment date. On enrollment day (if not already done), finish **before Shift 1 hits the 30-minute expiry rule**.
- If Week 0 is still missing when that enrollment-day deadline passes → institute **manually edits enrollment date** to the next day; sessions stay blocked until Week 0 is submitted.
- Week 0 steps come from `weeklyFormDefinitions` for week `0` (may be fewer steps than later weeks).
- Week 0 is the **lock point** for patient edits — after submit, all patient fields (including enrollment date and Group Type) are **view-only**. Correct enrollment date must be set **before** early Week 0 submit.
- Sessions cannot start before `enrollmentDate`, even if Week 0 was submitted early.

6.4 Weekly form ordering rules

- Forms only for **even weeks** listed in `weeklyFormDefinitions` (`0, 2, 4, 6, 8…` as configured — count is independent of `trialDurationWeeks`).
- Cannot submit Week N while the previous even week (N−2) is still **open and not submitted**.
- If the previous even week's window has **expired (missed)**, the next even week in config is unblocked automatically.
- Institute cannot skip backwards (cannot submit Week 2 after Week 4 is already submitted).
- Once a weekly form is fully completed, it **cannot be edited** from the mobile app. View-only access is available from the Trial Admin Panel.

6.8 Weekly Form PIN (access control)

To prevent unauthorised weekly form submissions, a **6-digit numeric PIN** is required before the form can be opened.

How it works:

```
Institute staff taps "Weekly → Start" on Dashboard
    ↓
PIN entry screen shown
    ↓
POST /weekly-assessments/verify-pin  { pin, institutionId }
    ↓ (valid PIN)
Backend returns short-lived token (valid 1 hour)
    ↓
Weekly form Steps (configured for that week) → Submit non-HRV steps (token in request header)
    ↓
HRV step if configured (5-Minute Resting HRV) → Submit (token in request header)
```

PIN lifecycle:

Detail	Value
Format	6-digit numeric
Scope	One active PIN per institute at a time
Generated by	Cron job — every **14 days** (aligned with `weeklyFormIntervalDays`)
Stored in	Trial DB — hashed, one active PIN per institute at a time
Communicated to institute	Manually by Skitii admin via SMS / call
Token validity after verify	**1 hour**
Wrong PIN lockout	After **10 attempts** (configurable via env variable)
Lockout reset	**Automatically after 30 minutes**
PIN issue resolution	Institute contacts Skitii admin directly (phone/SMS) — no admin panel action needed

Verification logic:

- Backend checks if the submitted PIN matches the **current active PIN** for that institute — no `weekNumber` validation in the API.
- PIN rotates every **14 days** purely for security so it cannot be reused indefinitely.

Week 0 PIN:

- Week 0 form also requires a valid PIN. The first PIN is generated before or at enrollment start for that institute (then rotates every 14 days).

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8. Dashboard — Action Button State Machine

The backend returns session/form state params per shift per patient. The frontend renders buttons accordingly:

Session status	`terminationReason`	Post Therapy form	Shift window	Trial status	Button(s) shown
Not started	—	—	Open	Active	**Start**
Not started	—	—	Closed	Active	**Missed** (no action)
Not started	—	—	Open	Completed (after `completionDate`)	**—** (no button)
`INCOMPLETE`	`USER_ENDED`	Not submitted	Open	Active	**Restart** + **Resume**
`COMPLETED`	`SESSION_MAX_REACHED`	Not submitted	Open	Active	**Resume**
`AUTO_TERMINATED`	`TIMEOUT` or `AUTO_TERMINATED_ON_NEW_SESSION`	Not submitted	Open	Active	**Restart** + **Resume**
`INCOMPLETE`	`USER_ENDED`	Submitted	Open	Active	**Restart**
`COMPLETED`	`SESSION_MAX_REACHED`	Submitted	Open	Active	**Done**
`AUTO_TERMINATED`	`TIMEOUT` or `AUTO_TERMINATED_ON_NEW_SESSION`	Submitted	Open	Active	**Restart**

Restart: Creates a **new session** for that shift within the open window. Previous session data is **kept** in DB.

Resume: Opens the Post Therapy form for the existing session.

Backend response params (per shift, per patient in list API):

```
{
  "shift": "shift1 | shift2 | shift3",
  "shiftStatus": "upcoming | active | missed",
  "sessionStatus": "not_started | ACTIVE | INCOMPLETE | COMPLETED | AUTO_TERMINATED",
  "terminationReason": "USER_ENDED | SESSION_MAX_REACHED | TIMEOUT | AUTO_TERMINATED_ON_NEW_SESSION | null",
  "postFormStatus": "not_submitted | submitted",
  "trialStatus": "active | completed",
  "actionButtons": ["start" | "restart" | "resume" | "done" | "missed"],
  "weekly": {
    "buttonLabel": "Week 0 | Week 2 | Week 4",
```

```
    "currentStudyWeek": 0,
    "status": "pending | done | missed",
    "weeklyFormId": "optional form id when steps already submitted"
  }
}
```

> `weekly.buttonLabel` drives the Dashboard Weekly chip text. Form steps come from `getInstitutionTrial.weeklyFormDefinitions` matched by `currentStudyWeek` — not from the dashboard chip.

---

9. Dashboard — Patient List

8.1 Sort order

- **Default sort:** Patients with pending sessions first (Shift 1 → Shift 2 → Shift 3), then completed, then missed last.
- **Optional sorts:** Compliance (low → high), Enrollment Date.

8.2 Columns

Column	Source
Name & Registration No.	Patient record (`registrationNo` — optional, same as Skitii product)
Age	Patient record
Action (shift chips)	Backend shift state per patient
Group Type	Patient record (Adaptive / Control)
Weekly (button label / Done)	Backend `weekly.buttonLabel` + status (even weeks only)
Enrollment Date	Patient record
Treatment Duration	From trial config (`trialDurationWeeks` — Yenepoya **4 Weeks**)
Compliance Status	Computed: completedSessions / expectedSessions
Emergency Number	Patient record
Last Session	Last completed Post Therapy `completedAt`

8.3 Filters

- **Date filter** — defaults to today; can select a specific date to view that day's session status.
- **Shift chips:** `All` | `Shift 1` | `Shift 2` | `Shift 3` — filters by session status for the selected shift on the selected date.
- **Session status chips:** `All` | `Pending` | `Completed` | `Missed` — filters by session status for the selected date + shift.
- **Weekly form chip:** `Weekly Pending` | `Weekly Submitted` — filters by current week's form status.
- **Search** — by patient name only.

- **Pagination** — server-side.

---

10. Add Patient (Enrollment)

9.1 Form sections

Personal Information

- First Name*, Last Name*
- Patient ID / Registration No. (`registrationNo` — optional)
- Emergency Number* (India +91)
- Language Preference* — Kannada | English | Hindi | Malayalam
- Age* (years, integer)
- Gender* — Male | Female | Other

Clinical Data *(Yenepoya v1 — will differ per future hospital)*

- Diagnosis Stage* — Stage 1 | Stage 2 | Stage 3 | Stage 4
- Treatment Type* — Radiotherapy | Chemotherapy | Both
- Cancer Site* (text — specific Head & Neck location)
- ECOG Performance Status* — 0 | 1 | 2 | 3 | 4

Study Enrollment

- Enrollment Date* (date picker, dd/mm/yyyy)
- Surgical Type* — Pre-Operative | Post-Operative
- Group Type* — Adaptive | Control

> **Expected Duration** is **not** on this form. It is set on trial config (`trialDurationWeeks` = **4** for Yenepoya) and applies to all patients.

CTA: **Enrol Patient** (disabled until all fields valid)

9.2 Rules

- Patient is **editable** until Week 0 form is submitted (including **enrollment date** and group type). If Week 0 will be submitted early, set enrollment date first.
- After Week 0 submission → **view only** (no edits from the mobile app). View-only access is available from the Trial Admin Panel.
- **Group type (Adaptive / Control)** can be changed only **until Week 0 is submitted**. Blocked after that.
- **Registration No.** (`registrationNo`) — optional at enrollment, same field as the Skitii product. If provided and duplicate within the institute, the backend rejects it with a validation error (same logic as the existing Skitii product backend).
- Enrollment spanning a month boundary uses normal date arithmetic — no special handling required.
- Future hospitals: patient intake form will differ; backend must support per-institution schema versioning.

---

11. Statistics

Shown per institute (Yenepoya). **Date range filter is mandatory** — options:

Option	Description
Today	Default
Last 3 Days	
Last 7 Days	
Last 2 Weeks	
Custom Date Range	User picks start and end date

4 KPI blocks displayed on the Statistics screen (all respect the date range filter):

Block	Label	Definition
1	**Total Sessions**	Expected patient × shift slots in range (Week 0 submitted, within enrollment–completion), broken down by Shift 1 / 2 / 3
2	**Total Completed Sessions**	Slots with Post Therapy submitted, broken down by Shift 1 / 2 / 3
3	**Total Missed Sessions**	Slots where the shift window closed with no Post Therapy submitted, broken down by Shift 1 / 2 / 3
4	**Registered Patients**	Patients whose `createdAt` falls in the selected clinical date range

> **Completed session = Post Therapy form submitted. Missed session = shift window closed + no Post Therapy submitted for that shift.** On a fully closed day: Total Sessions = Completed + Missed. Open/upcoming shifts today count in Total only.

---

12. Auth & PIN

11.1 Login flow

```
Login screen (email input)
    ↓ sendOtp (Existing Auth Backend)
OTP screen
    ↓ verifyOtp → JWT
Create PIN (if not set on this device)
    OR
Enter PIN (if already set)
    ↓
Dashboard
```

- PIN is **4 digits**, stored in device Keychain/Keystore only (not on any server).
- PIN is **per-device** — each tablet has its own PIN for the same institute account.

- No Administrator role in the trial app — all logged-in institute users have equal access.

11.2 Forgot PIN

1. User taps **Forgot PIN?** on Enter PIN screen.
2. Trial Backend sends OTP to **institute's login email** (device-specific reset request).
3. User enters OTP on tablet.
4. On verify → PIN cleared on that device only → user is taken to **Create PIN** screen.
5. This resets the PIN **for that device only**. Other tablets are unaffected.

---

13. Control Music — Admin Setup

1. In **Skitii Admin Frontend**, admin adds a flag `controlPatientMusic: true` to music assets.
2. These flagged tracks form the **control music pool**.
3. The `GET /music` API for control patients returns only flagged tracks.
4. On session start for a control patient: frontend picks one **random track** from the pool and plays it continuously.
5. If the patient changes music: another random track from the pool is selected and plays continuously.
6. Music **ranking is not updated** for control patients.
7. Session readings store the full music history per shift: `[{ trackId, startedAt, endedAt }]`.

---

14. Trial Admin Panel

A new dedicated web admin panel built for managing the Skitii Trial. Separate from the existing Skitii Admin Frontend.

13.1 Responsibilities

Module	Description
Patient Management	View all enrolled patients, their details, group type (Adaptive / Control), enrollment date, and status
Session Data	View per-patient session history — Pre Therapy, Baseline, Live Session, Post Therapy data per shift per day
Weekly Assessments	View submitted weekly form responses (dynamic steps per even week; 5-min HRV when present)
Statistics Dashboard	Admin-level overview of all KPIs across the trial (sessions, missed sessions, compliance, weekly reports)
Data Export	Export trial data (patients, sessions, assessments) as CSV or PDF for clinical analysis
Institute Management	Add and manage hospital institutes and their trial assignments (supports future multi-hospital trials)
Trial Configuration	Configure trial parameters per institute (duration, patient intake schema, email reminders)

13.2 Access

- Accessible by Skitii internal team only (not the institute/hospital).
- Authenticated via existing Auth Backend (separate admin role).

---

15. Trial Backend — API Summary

The Trial Backend serves both the **React Native Mobile App** (tablet) and the **Trial Admin Panel**. APIs listed here are for the mobile app; the Admin Panel will have additional endpoints for data management and exports.

API	Description
`POST /auth/forgot-pin/send-otp`	Send OTP to institute email for PIN reset (device-specific)
`POST /auth/forgot-pin/verify-otp`	Verify OTP, clear PIN for that device
`POST /patients`	Enroll new patient
`GET /patients`	List patients with shift state, weekly payload (`buttonLabel` , `steps` , `availableWeeks`), compliance
`GET /patients/:id`	View patient detail
`PATCH /patients/:id`	Update patient (blocked after Week 0 submitted; enrollment date editable until Week 0 submitted)
`POST /sessions/start`	Start session — validates: clinical date ≥ `enrollmentDate` (`BEFORE_ENROLLMENT`), Week 0 submitted (`WEEK0_REQUIRED`), shift window, shift ordering, duplicate session, 30-min minimum; later weekly forms do not gate sessions; allowed through `completionDate` inclusive (`TRIAL_COMPLETED` after)
`POST /sessions/validate`	Safety-net check after Pre Therapy form — re-validates 30-min minimum before proceeding to Baseline
`POST /sessions/:id/stop`	Mark session as `INCOMPLETE` / `USER_ENDED`
`POST /sessions/:id/complete`	Mark session as `COMPLETED` / `SESSION_MAX_REACHED` (20-min auto-end from client)
`POST /sessions/:id/readings`	Upload gzip readings chunk (Adaptive: per shift; Control: per music segment)
`POST /sessions/:id/post-therapy`	Submit Post Therapy form
`POST /weekly-assessments/verify-pin`	Verify weekly form PIN — returns short-lived token (1 hr) on success; lockout after 10 wrong attempts
`POST /weekly-assessments`	Submit non-HRV steps for the active even week (batched) — requires PIN token; steps validated against `weeklyFormDefinitions`
`POST /weekly-assessments/:id/hrv`	Submit 5-minute HRV step when configured — requires PIN token
`GET /statistics`	Fetch all KPI aggregates (filter: dateRange)
`GET /music/control`	Fetch control patient music pool

16. Edge Cases & Error Handling

16.1 Enrollment

Scenario	Behaviour
Patient enrolled on last day of month (e.g. Jul 31)	`completionDate` = enrollment + (`trialDurationWeeks` × 7 − 1) days (Yenepoya +27); spans months normally
Duplicate `registrationNo` entered for two patients (when provided)	Same validation logic as existing Skitii product backend — rejected with error
Group type change after enrollment	Allowed **until Week 0 form is submitted**. Blocked after that.
Week 0 submitted early (before enrollment date)	Allowed; patient locked view-only. Fix enrollment date **before** this submit
Sessions before `enrollmentDate`	Not allowed — `BEFORE_ENROLLMENT` (even if Week 0 already submitted)

16.2 Session start — two-point validation

When the user taps **Start** on the Dashboard, `POST /sessions/start` validates both conditions in a single API call:

Check	Error code	Frontend action
Clinical date before `enrollmentDate`	`BEFORE_ENROLLMENT`	Start disabled — sessions only from enrollment day
Week 0 not submitted	`WEEK0_REQUIRED`	Start disabled until Week 0 done
Trial past `completionDate` (next calendar day+)	`TRIAL_COMPLETED`	Start disabled; sessions blocked (weekly forms still allowed if window open)
Duplicate session (same device, same shift)	`SESSION_ALREADY_ACTIVE`	Same redirect/popup as Skitii product
Duplicate session (different device, same shift)	`SESSION_ACTIVE_OTHER_DEVICE`	Show confirmation popup — user must confirm to proceed
Less than 30 minutes remain in shift window	`SHIFT_WINDOW_EXPIRING`	Show error, block start
Shift window already closed	`SHIFT_WINDOW_CLOSED`	Start button already disabled; backend returns error as safety net

Second check (safety net) — after Pre Therapy form submit:

Frontend calls the API again after the Pre Therapy form is filled to re-validate the 30-minute minimum. This catches cases where the shift window was fine at Start but the user spent too long filling the form.

Check	Error code	Frontend action
Less than 30 minutes remain now	`SHIFT_WINDOW_EXPIRING`	Show error, cannot proceed to Baseline

16.3 Mid-session events

Scenario	Behaviour
BLE device disconnects mid-session	Same popup/redirect behaviour as Skitii product frontend
App crashes mid-readings upload	Same recovery mechanism as Skitii product
Session runs past ~20 minutes	Client auto-end → `COMPLETED` / `SESSION_MAX_REACHED`. Backend 30-min job → `AUTO_TERMINATED` / `TIMEOUT` if still `ACTIVE`.
New session started on same device while one is `ACTIVE`	Previous session → `AUTO_TERMINATED` / `AUTO_TERMINATED_ON_NEW_SESSION` (same as Skitii product)

16.4 Weekly forms

Scenario	Behaviour
Week 0 never submitted	All sessions blocked — `WEEK0_REQUIRED` until Week 0 is submitted
Week 0 submitted before `enrollmentDate`	Allowed. Patient becomes view-only. Sessions still blocked until enrollment day (`BEFORE_ENROLLMENT` if Start tapped early)
Attempt to start session before `enrollmentDate`	`BEFORE_ENROLLMENT` — Start disabled; sessions only from enrollment date onward
Week 2 / 4 / 6… pending or missed	Sessions still allowed (from enrollment through `completionDate` inclusive). Only Week 0 + enrollment date gate sessions
Week 0 still missing when Shift 1 30-min rule hits on enrollment day	Institute manually edits enrollment date to the next day (no backend auto-change); retry Week 0; sessions stay blocked until Week 0 is submitted
Enrollment date edited before Week 0 submitted	Allowed; `completionDate` and even-week form windows recalculated. Blocked after Week 0 submitted (set date before early Week 0)
Even-week form window (Week 2+)	`windowStart` = enrollment + (week × 7); `windowEnd` = start + 6 days (7-day inclusive window). e.g. enroll 17 Jul → Week 2 = **31 Jul – 6 Aug**
Form weeks after `completionDate` (e.g. Week 4 / 6 / 8)	No sessions for those slices after completion; forms still open on schedule and submittable while window open
Previous even week missed (window expired, not submitted)	Next even week in `weeklyFormDefinitions` becomes available on its scheduled window
Submit Week N while previous even week (N−2) is still open and not submitted	Rejected — must submit or miss previous even week first
Submit Week N after a later even week already submitted	Rejected — cannot skip backwards
Odd week requested (1, 3, 5…)	Rejected — forms only for even weeks in config
Week not in `weeklyFormDefinitions`	Not shown on dashboard; submit rejected (duration does **not** decide this)
Form window not open yet / already expired	Submit rejected; dashboard shows pending / missed accordingly
Weekly form after `completionDate`	**Allowed** if week is in `weeklyFormDefinitions` and form window is still open. Sessions remain blocked from the day after completion
More form weeks than `trialDurationWeeks` (e.g. duration 4, **7** form entries)	Valid. Sessions through completion (+27 for Yenepoya); remaining form weeks fillable while windows open
Duration 3 weeks (`completionDate` = enrollment + 20) with forms after that	Sessions blocked from day after completion; configured forms still OK if window open

Scenario	Behaviour
Form reminder day (`windowEnd − 1`)	Email only that day; **skip** if form already Done; **no** Week 0 reminder
Form reminder on other days of the window	No email (not daily; not Thu/Fri/Sat)
Step not in that week's configured `steps`	Rejected — payload must match `weeklyFormDefinitions` / `stepsConfigured`
Progress UI	Show **Step X of N** where N = steps configured for that week only (not full catalogue size)
`treatment_adherence` outside weeks where configured (Yenepoya: Week 4)	Step not rendered; submit rejected if sent
`satisfaction_survey` for control patient, or outside configured intervention week	Step hidden / rejected — intervention (Adaptive) group only
Non-HRV steps submitted, HRV still pending	Status `steps_submitted`; week not Done until HRV posted when `hrv` is configured
HRV submit with expired PIN token	`WEEKLY_PIN_TOKEN_EXPIRED`; re-verify PIN, then resubmit HRV
SpO_2 / BP out of "clinical" range	Accepted — **no min/max range validation** (fields still required)
Form edited after submission	Not allowed from mobile. View-only from Trial Admin Panel
Wrong PIN entered	Attempt counter incremented; error shown
Wrong PIN 10 times (configurable)	Locked 30 minutes; `WEEKLY_PIN_LOCKED`
PIN token expires (1 hour)	Submission rejected with `WEEKLY_PIN_TOKEN_EXPIRED`; re-verify PIN
PIN entered on tablet B while tablet A has active token	Both tokens valid independently
PIN rotation every 14 days (not Sunday night)	Old PIN deactivated; institute must use new PIN (admin informs via SMS/call)
Institute not informed of new PIN	Institute contacts Skitii admin by phone/SMS — no system action

16.5 Calendar edge cases

Scenario	Behaviour
Patient enrolled on a Sunday	Week 0 due by that day (may submit earlier); Shift 1 30-min rule if submitting Sunday
Form window weekday	Follows enrollment weekday + (week × 7); always **7 days** inclusive (not forced Mon–Sun)
Enrollment spans month boundary	No special handling — date arithmetic applies normally (`enrollment + trialDurationWeeks × 7 − 1`)
Enrollment date manually moved forward (before Week 0 submitted)	`completionDate` and form windows recalculated from new enrollment date
Form weeks configured beyond duration calendar	Form windows from enrollment + (week × 7); session stop date unchanged

16.6 After completion date

Scenario	Behaviour
Patient tries to start session on `completionDate`	**Allowed** (sessions through completion day inclusive)
Patient tries to start session from the next calendar day	Backend returns `TRIAL_COMPLETED` ; Start button disabled
Weekly form submitted after completion date	**Allowed** if week is in `weeklyFormDefinitions` and form window is still open
Yenepoya: enroll 17 Jul, completion **13 Aug**, Week 4 form **14–20 Aug**	Sessions through 13 Aug; Week 4 form OK after; reminder **19 Aug** if still pending
Yenepoya duration 4 with form weeks 6 / 8...	Sessions through +27 days; later form weeks remain fillable while each window is open

17. Key Technical Constraints

1. **Tablet-only** (landscape, **Android**). No phone layout.
2. **Any compatible BLE heart rate device** — not locked to a specific model. Tablet reads RR intervals; HRV computed on-device.
3. **On-device compute** — RMSSD, SDNN, pNN50 computed on tablet, uploaded to Trial Backend (same as product pattern).
4. **Always online** — tablets have internet. All Trial APIs (sessions, weekly forms, dashboard) require network. No offline queue.
5. **Concurrent multi-tablet** — Trial Backend must block a second session for the same `patientId` + `shift` + `date` if one is already `ACTIVE` .
6. **Shift 3 date attribution** — a session starting at 2AM July 18 is stored as Shift 3 of July 17 (`sessionDate: 2026-07-17` , `shift: 3`).

7. **Future multi-hospital** — patient intake schema must be versioned per `institutionId` / `trialId` to support different fields for other hospitals.
8. **Weekly Form PIN** — PIN is hashed in DB, never stored in plaintext. Rotates every **14 days**. Lockout counter and token are stored server-side (not on device).